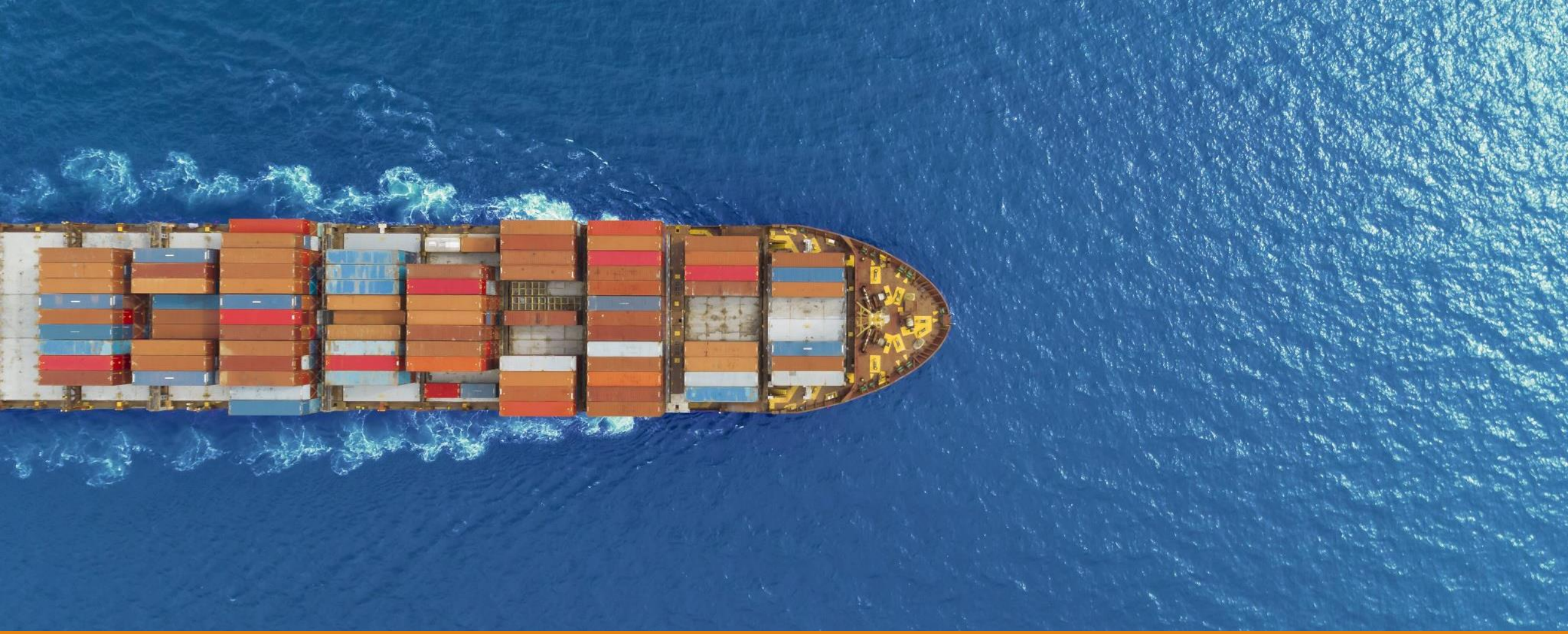


Data as Capital in TAI Economies
A Biomorphic Framework

BY BELLA SMITH

Why Traditional Economics Falls Short

Data violates core assumptions of traditional capital theory:

- Non-rival (unlimited simultaneous use)
- Self-generating (can create more of itself)
- Appreciates rather than depreciates in many contexts
- Value exists in superposition until "observed" through application

TAI magnifies these unique properties dramatically.

Data as Both Capital and Living Organism

Economic Capital	Biological Organism
Productive asset	Evolving entity
Production input	Self-replicating structure
Market commodity	Ecological participant
Static value	Adapting characteristics

Four Pillars of This Approach

Extended Capital Theory: New production functions incorporating data's unique properties

Evolutionary Dynamics: How data evolves through mutation and selection in TAI environments

Quantum-Inspired Valuation: Resolving the paradox of data's multiple potential values

Symbiotic Systems Theory: Modelling human-TAI-data interdependencies

Data as Living Digital Organisms

Data structures contain "digital DNA" determining function and fitness

TAI systems act as both environment (selection pressure) and mechanism (mutation)

Some data ecosystems demonstrate self-sustaining properties (autopoiesis)

Quantum-Inspired Valuation

Data exists in superposition of multiple potential values

Value "collapses" into specific states when applied in particular contexts

Data elements become "entangled" when processed together

Observer effects: The act of valuing data changes its characteristics

Policy and Governance Recommendations

Context-sensitive property rights regimes vs. one-size-fits-all ownership

Adaptive governance with built-in feedback mechanisms

Multi-stakeholder governance models beyond markets or states

Novel approaches to valuation, taxation, and benefit distribution

Next Step: The Human Dataome Project

Mapping the totality of human-related data and its economic impacts

Tracking evolutionary lineages of data structures

Identifying critical dependencies and ecosystem relationships

Anticipating phase transitions in data economies

Conclusion

Data represents something genuinely novel in economic history

Blurs boundaries between capital, information, and autonomous agent

Framework enables more accurate prediction and more effective governance

Provides foundation for ensuring TAI benefits are broadly shared